C6.3 Issuing snoop transactions

It is the responsibility of the interconnect to generate the snoop transactions required to progress a transaction from an initiating master.

The transaction from the initiating master determines which cached masters in the shareability domain must be snooped:

- The following transactions do not cause a snoop of any cached masters:
  - ReadNoSnoop
  - WriteNoSnoop
  - WriteBack
  - WriteClean
  - Evict.
- The following transactions must cause a snoop of the cached masters that can hold a copy of the cache line:
  - ReadOnce
  - ReadClean
  - ReadNotSharedDirty
  - ReadShared.

  Snooping of the cached masters must continue until any one of the following occurs:
  - a copy of the line is obtained
  - a snoop response is received with WasUnique, **CRRESP[4]**, asserted
  - all caches have been snooped.
- The CleanShared transaction must cause a snoop of the cached masters that can hold a copy of the cache line until any one of the following occurs:
  - a dirty copy of the line is obtained, as indicated by snoop response PassDirty, **CRRESP[2]**, being asserted
  - a snoop response is received with WasUnique, **CRRESP[4]**, asserted
  - all caches have been snooped.
- The following transactions must cause a snoop of the cached masters that can hold a copy of the cache line:
  - ReadUnique
  - CleanUnique
  - MakeUnique
  - CleanInvalid
  - MakeInvalid
  - WriteUnique
  - WriteLineUnique.

  Snooping of the cached masters must continue until either of the following occurs:
  - a snoop response is received with WasUnique, **CRRESP[4]**, asserted
  - all caches have been snooped.

Note

The interconnect must not issue a snoop transaction to the initiating master. 不能对请求发起者发送Snoop请求

Table C6-1 on page C6-230 shows for each transaction issued by the initiating master:

- the snooped cache line state change that must be ensured by the interconnect
- the snoop transaction that this specification recommends the interconnect to use
- the optional snoop transactions that the interconnect can use.

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In Table C6-1, the snoop transaction that this specification recommends is also included as an optional snoop transaction.

Table C6-1 Interconnect snoop requirements

Transaction from Initiating Master	State change for the snooped cache	Recommended Snoop transaction	Optional Snoop transaction
ReadNoSnoop	None	-	-
ReadOnce	None	ReadOnce	ReadOnce ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid CleanShared
ReadClean	Shared or Invalid	ReadClean	ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid
ReadNotSharedDirty	Shared or Invalid	ReadNotSharedDirty	ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid
ReadShared	Shared or Invalid	ReadShared	ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid
ReadUnique	Invalid	ReadUnique	ReadUnique, CleanInvalid[a]
CleanUnique	Invalid	CleanInvalid	ReadUnique, CleanInvalid[a]
MakeUnique	Invalid	MakeInvalid	ReadUnique, CleanInvalid MakeInvalid
CleanShared	Clean or Invalid	CleanShared	ReadUnique, CleanInvalid CleanShared
CleanInvalid	Invalid	CleanInvalid	ReadUnique, CleanInvalid[a]
MakeInvalid	Invalid	MakeInvalid	ReadUnique, CleanInvalid MakeInvalid
WriteNoSnoop	None	-	-
WriteUnique	Invalid	CleanInvalid	ReadUnique, CleanInvalid[a]
WriteLineUnique	Invalid	MakeInvalid	ReadUnique, CleanInvalid MakeInvalid
WriteBack	None	-	-
WriteClean	None	-	-
Evict	None	-	-

a. Other optional snoop transactions can be used if the cached masters in the same shareability domain are not all snooped at the same time.

The interconnect is not required to snoop all caches at the same time, caches can be snooped sequentially.

When the interconnect is snooping multiple cached masters, it is not required to snoop all the cached masters in an identical manner.

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If the interconnect is carrying out the snoop transactions sequentially, that is, it is issuing some snoop transactions after other snoop transactions for the same cache line have completed, then after a snoop response is received with PassDirty asserted, it is permitted to use the MakeInvalid snoop transaction for the remaining cached masters that are still to be snooped. The transactions that can benefit from this use of the MakeInvalid snoop transaction are:

- WriteUnique
- ReadUnique ? ? ?
- CleanUnique
- CleanInvalid.

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